KNEE OSTEOARTHRITIS PREDICTION AND PROGRESSION USING MULTI-MODAL DEEP LEARNING.

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DECLARATION

I hereby declare that this dissertation is entirely my own work and has not been submitted, in whole or in part, to any other university or institute of higher learning for the award of any degree or diploma. To the best of my knowledge and belief, it does not contain any material previously published or written by another person, except where due acknowledgment has been made in the text.

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ACKNOWLEDGEMENT

I wish to extend my heartfelt thanks to my dissertation advisor, Ms. Jenny Krishara, and my co-supervisors, Ms. Thamali Dassanayake and Ms. Wishalya Tissera, for the tremendous advice and support extended to me during the duration of this study. The expertise, motivation, and guidance provided by them have helped in developing this dissertation to perfection.

I wish to thank the Faculty of Computing for extending the much-needed academic support to me during this study period. The valuable insights provided by them were very helpful in completing this dissertation successfully.

I wish to thank my colleagues for their cooperation, discussions, and sharing of ideas with me during the process of this study.

Lastly, I would like to thank all those who assisted me either directly or indirectly in the process of this dissertation.

ABSTRACT

Knee osteoarthritis (KOA) is a progressive joint disorder that greatly impacts the mobility and quality of life of individuals, particularly older patients. In the conventional diagnosis process of this condition, radiographic imaging and expert clinical judgment play a major role, which may be subjective, costly, and confined to one-time assessments. To overcome these challenges, this module suggests a multicomponent decision support system to automatically predict the severity of KOA and provide treatment recommendations.

The image component employs the EfficientNetB0 architecture, a deep-learning algorithm, for feature extraction from the knee radiographs, whereas structured clinical information is processed using the XGBoost machine learning model. In the late fusion approach, predictions made by both modalities are combined to increase their robustness and predictive accuracy. Moreover, we have developed a Treatment Management Support System using a set of rules that would enable making recommendations regarding the treatment level, follow-up, and necessary interventions based on the predicted KL grade as well as patient characteristics such as body mass index (BMI), symptoms, and clinical markers.

In conclusion, the suggested solution allows improving the accuracy, interpretability, and practical application of KOA severity predictions.

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LIST OF ABBREVIATIONS

Abbreviation	Description
OA	Osteoarthritis
KOA	Knee Osteoarthritis
MRI	Magnetic Resonance Imaging
BMI	Body Mass Index
AI	Artificial Intelligence
KL	Kellgren Lawrence
XAI	Explainable Artificial Intelligence
CNN	Convolutional Neural Network
MLP	Multi Layer Perceptron
CDI	Cartilage Damage Index
TF	TensorFlow

1. INTRODUCTION

Osteoarthritis is one of the most common degenerative joint disorders that affect many people worldwide. According to research findings, approximately 240 million people around the world have symptoms of OA. In addition, about 10% of men and 18% of women aged 60 years and above are suffering from OA [6]. In Sri Lanka, KOA affects a huge percentage of the adult population, especially females aged 50 years and above, as about 21.8% are diagnosed with the disease, while another 29.9% have severe or moderately severe forms of KOA [7].

KOA is a type of degenerative disorder that involves the gradual breakdown of cartilage, resulting in symptoms such as joint pain, swelling, stiffness, and limited movement ability. This disorder may progress into a point where KOA causes severe degeneration of the bones, ligaments, joint membrane, and other tissues. It eventually leads to joint deformities and long-term disability.

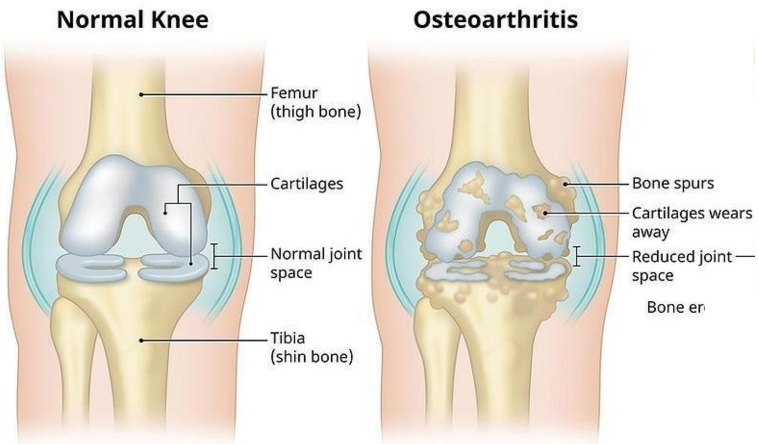

Figure 1– Normal Knee vs Osteoarthritis Knee

KOA severity can be classified based on the Kellgren-Lawrence grading scale that ranges from 0 to 4 where Grade 0 denotes normal state while Grade 4 stands for severe disease development. Early KOA stages can include absence or lack of symptoms while minimal structural alterations such as osteophyte formation may be observed. At moderate stages of the disorder, patients experience higher levels of pain and joint deformation necessitating medical treatment including physiotherapy and medication. In advanced KOA stages, joint space narrowing and serious cartilage degeneration occurs resulting in pain and the necessity of surgery, such as a knee replacement [8].

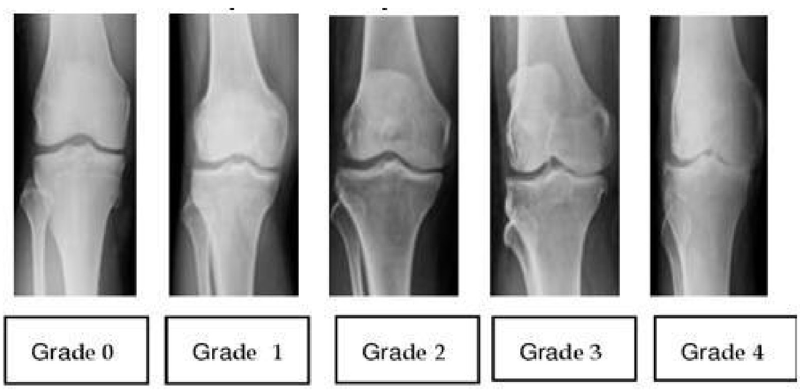

Figure 2 – Kellgren Lawrence scale

While the Kellgren-Lawrence grading system provides an opportunity for proper classification, present diagnostic methods involve radiological assessment of bone changes, which is expensive. Also, these techniques are rather difficult to conduct, which means special equipment and highly trained personnel is required. Moreover, they are limited by the fact that these techniques only allow making single-point assessments that do not reflect progressive development of the disease.

In light of these restrictions, the demand for computerized, objective, and scalable solutions to effectively determine the severity of KOA is increasingly recognized. This will not only improve the process of diagnosis but also help in identifying and planning treatments as well as managing patients more effectively in the long term. Considering this situation, the application of artificial intelligence approaches on multimodal data becomes quite significant.

1.1. Background Literature

Knee Osteoarthritis has emerged as one of the significant healthcare challenges in today's world, prompting research in automated diagnosis, severity prediction, and decision-making assistance through artificial intelligence applications. Various studies have been conducted to investigate different solutions, ranging from image-based deep learning methods to clinical data-based machine learning algorithms and even multi-modal systems that integrate several types of input.

Image-based methods have proven to be efficient in detecting and assessing KOA cases. For instance, Wang et al. (2022) used deep learning models to classify KOA cases using MRI images, confirming the effectiveness of transferring learning architectures to analyze complex anatomical structures with high classification accuracy [7]. This study emphasizes the ability of deep neural networks in capturing minute changes in tissues which are difficult to detect by humans' naked eyes. On the other hand, Mehta and Kaur (2025) introduced an image-based deep learning solution consisting of CNN-attention mechanism to grade the severity of KOA based on X-ray images, resulting in improved accuracy [5].

Current advances have involved the application of various types of data in improving prediction performance. According to Chen et al. (2023), a recent advance involves a bimodal framework that incorporates both clinical data and image data in predicting the severity of KOA patients [3]. This indicates that it is beneficial to consider different data sources that help provide a more holistic assessment of patients' conditions.

In addition, Rameez et al. (2024) proposed an integrated system for detecting KOA from images and providing treatments through machine learning methods [2]. The study shows how possible the combination of diagnosis and treatment recommendation can be. However, the system is mainly based on image analysis and does not involve extensive clinical and biomarker data.

Nevertheless, current works focus more on enhancing the prediction performance or considering one kind of data while overlooking their potential clinical significance and interpretability. For example, it is rare to find a system that takes into account the integration of multimodal prediction and treatment recommendation. Consequently, the purpose of this research is to bridge such gaps through designing a new multimodal system for treating KOA.

1.2. Research Gap

KOA is a degenerative joint disease that demands early and precise diagnosis in order to avoid severe disability and enhance patient outcome. Although KOA research has advanced substantially, current grade severity systems are still poor, and fragmented. Existing studies can roughly be divided into two approaches:

Image based grading:

Deep learning models from knee radiographs or MRI images (e.g., Wang et al., 2022; Mehta & Kaur, 2025) have demonstrated encouraging accuracy. However, these approaches are limited to clinical settings, require expensive imaging instruments, and they only provide a static 'snap shot' of the disease. They do not support a continuous monitoring nor capture functional changes over time.

Clinical based assessment:

Other studies (e.g., Shidore & Jalnekar, 2023) use patient surveys, statistical risk models or epidemiological factors like age, BMI and occupation. While these provide useful information on population level risks they are subjective, they are retrospective and they are less precise than imaging methods.

This division shows that there is essentially a gap between the existing literature, which mainly employs either image-based or clinical based grading, and the recent advancements where the two are combined. Although attempts (e.g., Chen et al., 2023) have been made to fuse simple health metrics with imaging data, these are still restricted to single point predictions and miss the power of dynamic sensor-based biomechanical data.

Chen et al. (2023) - Predicting Severity of Knee Osteoarthritis Using Bimodal Data and Machine Learning

Mathevon and his colleagues developed a machine learning framework that integrates infrared thermographic images with patient health information (including BMI and age) and uses it to predict the severity of KOA. Their approach was to extract temperature and texture features from thermal images, combine them with clinical metrics and use the SMOTE technique to deal with class imbalance. Of the models tested, they found that a decision tree classifier was the most successful, with 85.71% accuracy, and concluded that bimodal data fusion substantially improves predictive performance over single mode strategies.

Wang et al. (2022) - Classification of Knee Osteoarthritis Based on Transfer Learning Model and Magnetic Resonance Images

They used the InceptionResNetV2 architecture pre-trained on ImageNet to derive features from knee MR images from the OAI-ZIB dataset. Their paper contrasted two optimization methods, SGD and RMSprop, using two data partitioning schemes. The results showed that the model with RMSProp was capable of high performance with up to 96.1% accuracy under image-level splitting, which proved the effectiveness of deep transfer learning for KOA MRI-based diagnosis.

Shidore & Jalnekar (2023) - Unveiling Risk Factors of Knee Osteoarthritis using Statistical and Machine Learning Model

Associations were analyzed using statistical methods including confidence intervals, odds ratios, and ch-square tests and machine learning models including SVM and Random Forest were used for knee pain prediction. They found that age, BMI, sedentary work and heavy physical activity were significant risk factors, and an SVM model predicted the presence of cardiovascular disease with 88% accuracy.

Mehta & Kaur (2025) - Automating Knee Osteoarthritis Severity Grading with CNN-Attention Hybrid Models

They suggested that a hybrid deep learning model (composed of CNN and attention mechanism) can be used to automate the grading of KOA severity from X-ray images using the KL scale. Their model was trained on publicly available datasets from OAI, and it used Grad-CAM visualizations to illuminate diagnostically relevant regions, such as joint space narrowing and osteophytes. The resulting system obtained an average accuracy of 92.0% with benefits not only for its high classification performance but also for its enhanced interpretability for clinical use.

This research fills these gaps by creating a multi modal KOA severity grading system that combines radiographic image features together with continuous biomarkers and clinical data. By incorporating imaging and force sensing modalities, the system will give a more complete picture of KOA progression. Crucially, it is for on device processing, so it can monitor in real-time without any need for continuous internet access.

By breaking away from static diagnosis to dynamic, patient entered management, this fusion based framework provides clinicians with more powerful decision support and provides patients with a means for ongoing, personalized monitoring in the clinic and in their everyday lives.

Feature / Aspect	**Wang et al. (2022) MRI**	**Mehta & Kaur (2025) X-ray**	**Shidore & Jalnekar (2023)Survey**	**Proposed System**
Primary Data Type	Medical Images (MRI)	Medical Images (X-ray)	Survey / Clinical Data	Multi-Modal (Clinical + Imaging)
Multi-Modal Data Fusion	No +	No +	No +	Yes
Measurement Scope / Environment	Clinical / Lab	Clinical / Lab	Field / Self-Reported	Yes
Continuous Monitoring	No +	No +	No +	Yes
Biomechanical Data	No +	No +	No +	Yes
AI / Analytics	No +	No +	Yes	Yes
Clinical/Patient Use	No +	No +	Yes	Yes
Internet Needed	No +	No +	Yes	Yes
Treatment Suggestions	No +	No +	No +	Yes

Table 1- Comparison of former research

1.3. Research Problem

Knee osteoarthritis is an insidious and progressive disease requiring prompt identification and proper management to avoid the development of severe disability and a decrease in quality of life. However, despite existing diagnostic procedures for the assessment of the severity of the condition, there are numerous problems associated with current practice in the evaluation of the disease. For instance, imaging modalities including radiography and MRI scan together with professional interpretation are commonly employed for identifying and characterizing disease progression. Nevertheless, this diagnostic procedure involves several drawbacks, including high costs, need for special equipment, subjectivity in severity grading, and inability to assess temporal dynamics.

Clinical and biomarker assessments try to compensate for the mentioned weaknesses by incorporating more personalized data into the examination procedure, including clinical features, history, or physiological parameters. Still, these procedures remain subjective and not precise enough to classify the severity grade correctly. Furthermore, they do not involve information obtained from imaging tests properly.

Another disadvantage of the technology is the absence of systems that assist with decisions besides predictive ones. The vast majority of existing models rely either on classification or grading, without offering suggestions about the treatment that could be provided following the results received from the algorithm. Consequently, there appears a gap between diagnostics and clinical decisions, rendering the usefulness of the technology questionable.

To overcome this issue, it would be necessary to develop a fully automated framework that would combine clinical and image data to enable a high-quality, precise and reliable prediction of KOA severity. Additionally, a developed model should close the gap between predicting and taking actions, providing suggestions regarding treatment based on severity grades and other clinical variables.

Thus, this paper tackles the challenge of creating an efficient, interpretable and clinically appropriate multi-modal decision support system for predicting the severity of knee osteoarthritis and managing its treatment using a rule-based approach.

1.4. Research Objectives

The primary goal of this module is to build a multimodal solution that would allow for the prediction of KOA severity with precision through radiography combined with other types of data like demographics, clinical data, and biomarkers. The solution is supposed to make it possible to perform classification on the basis of Kellgren–Lawrence (KL) grading scale. Additionally, this module aims at increasing its usability with the inclusion of a rule-based treatment management module for generating severity-driven recommendations.

1.4.1. Sub-Objectives

1. To build a predictive model for identifying KOA based on imaging data with deep learning methods (EfficientNetB0)
    - This objective is concerned with constructing a CNN model which can analyze images of knee X-rays to detect structural changes in the joint, including reduced joint space width, osteophyte formation, and cartilage erosion. Such models are trained on labeled data that helps classify severity of KOA based on the KL classification.

2. To construct a clinical predictive model with machine learning techniques (XGBoost)
    - It is necessary to create an ML model that uses structured clinical data (age, sex, disease manifestations, symptoms, and biochemical biomarkers). It will help to find non-linear relationships between features, which is a different approach to KOA identification compared to imaging methods.

3. To preprocess data for both imaging and clinical predictions
    - Such objective includes data pre-processing tasks for efficient model training. Preprocessing for imaging data implies resizing, converting images into grayscaled format, and reducing noise. Clinical preprocessing includes imputation and encoding tasks.

4. For the implementation of a multimodal fusion approach for enhanced prediction accuracy
    - The purpose here is to integrate the predictions of the two models, i.e., the imaging model and the clinical one, through a late fusion technique. In this way, by making use of multiple modalities of inputs, the system can be expected to have better accuracy than those working on a single modality.

5. To devise a severity level classification algorithm based on KL grade rating (KL0-KL4)
    - The system should be capable of classifying KOA into five distinct severity levels, allowing for consistent evaluation of the degree of advancement of the condition. This is important for consistent results in clinical settings.

6. To devise and develop a treatment recommendation rule-based system
    - This task requires development of an intelligent and interpretable system capable of generating suggestions for therapy depending on the severity level of KOA and individual characteristics like BMI, pain score, and symptoms. Outputs include therapy level, follow-up time, lifestyle advice, and clinical notes.

7. For the assessment of the system based on relevant metrics
    - In order to test the system, evaluation metrics like accuracy, precision, recall, f1 score, and confusion matrix are used to check the performance of the model in classifying and making reliable predictions based on severity.

8. For the incorporation of the system into an online web application
    - This goal centers around the implementation of the system into a web application, allowing it to be put to practical use by making real-time predictions and treatment suggestions for patients.

2. Methodology

2.1. Methodology

In here aims multi-modal Automated Knee Osteoarthritis (KOA) severity prediction and treatment planning by combining radiography images with clinical, biomarker, and demographic data. In particular, it provides an estimate for the Kellgren–Lawrence (KL) grading scale (KL0–KL4) and gives structured severity-aware recommendations with a rule-based treatment module. As compared with the initial idea, in the final version, machine learning is used instead of neural networks for clinical modeling, and there is a treatment module.

To build the imaging model, several deep learning approaches are utilized to identify radiological features in the knee X-ray image in grayscale. For the task, labeled data with different KL classes are available. Pre processing of the data set implies re sizing all X-rays to 224×224 and normalization. For better generalization, data augmentation is employed, namely random rotation and horizontal flips. Several Convolutional Neural Network (CNN) models, including DenseNet121, ResNet50, and EfficientNetB0, were investigated in terms of their effectiveness in the framework of transfer learning using pretrained weights of Imagenet. In comparison, the most suitable was the EfficientNetB0 architecture.

Concurrently, a clinical prediction pipeline analyzes structured data extracted from patient records, including demography (age, gender), anthropometrics (height, weight, BMI), and biochemistry (biomarkers). All personal patient information is anonymized to satisfy privacy requirements. The data pre-processing includes imputation of missing values via median imputation, encoding of categorical features, and feature normalization to guarantee numeric stability. Various types of machine learning models have been tried such as Logistic Regression, SVM, Random Forest and Gradient Boosting. However, the best results have been obtained by using XGBoost due to its high capability to deal with non-linearity and imbalance of the dataset. The resulting model provides outputs that correspond to KL severity prediction.

A late fusion method has been applied to exploit advantages of both imaging and clinical modalities simultaneously. The predictions of the EfficientNetB0 and the XGBoost models are combined in an ensemble decision tree-based algorithm. Such a way of fusion allows assigning higher weight to those predictions which are produced with more confidence.

On the basis of the predicted severity of the disease, we have designed a Treatment Management Support System based on rules. Different from previous techniques that adopted machine learning-based treatments, our system is designed based on rules extracted from orthopedic literature and expert opinions. Our rules take into account a number of patient-specific attributes such as predicted KL grade, BMI, pain, stiffness, swelling, and functional limitations. Our system is designed to produce treatment recommendation, follow-up time, lifestyle advice, and clinical notes. Cases with early-stage arthritis (KL0-KL1) receive advice for preventive treatment and lifestyle change, moderate cases (KL2-KL3) receive conservative treatment advice and physiotherapy, and severe cases receive medical intervention advice.

Our complete system is constructed based on a modular design, making it possible to independently update imaging model, clinical predictor, fusion technique, and treatment rules.

2.1.1. System Diagram

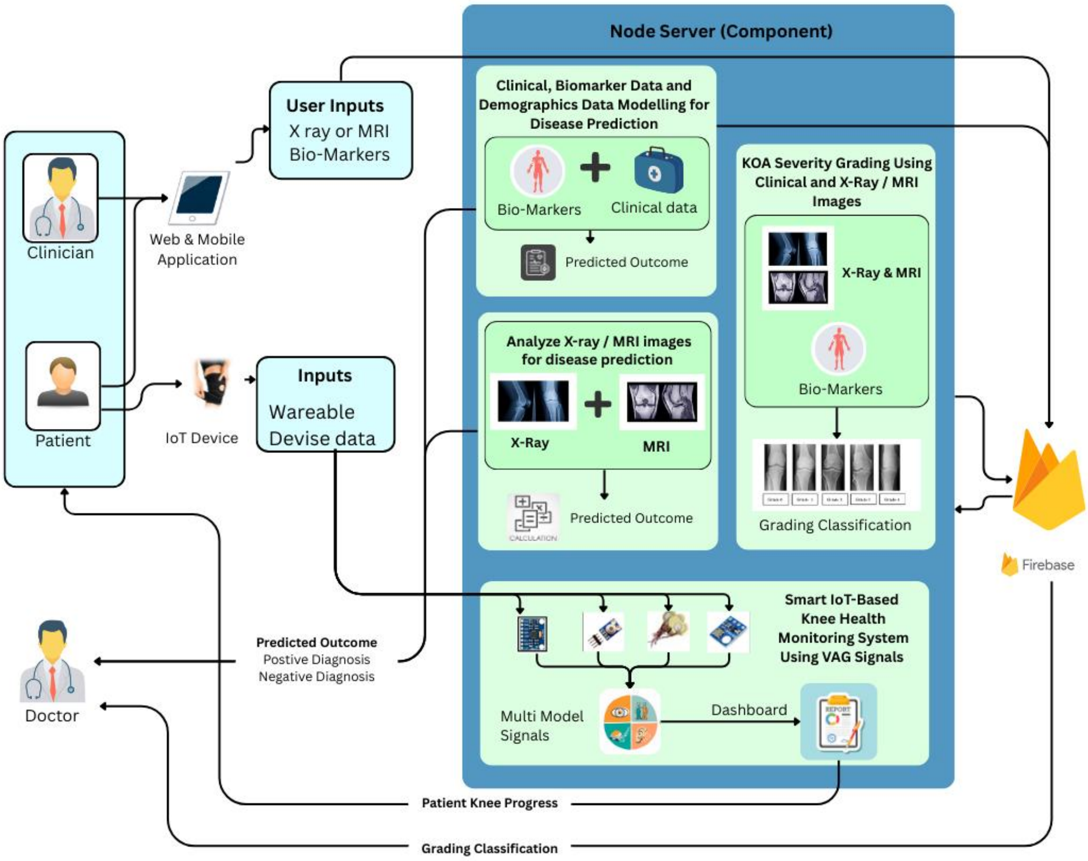

Figure 3 - Overall System Diagram

2.1.2. Component Architecture Diagram

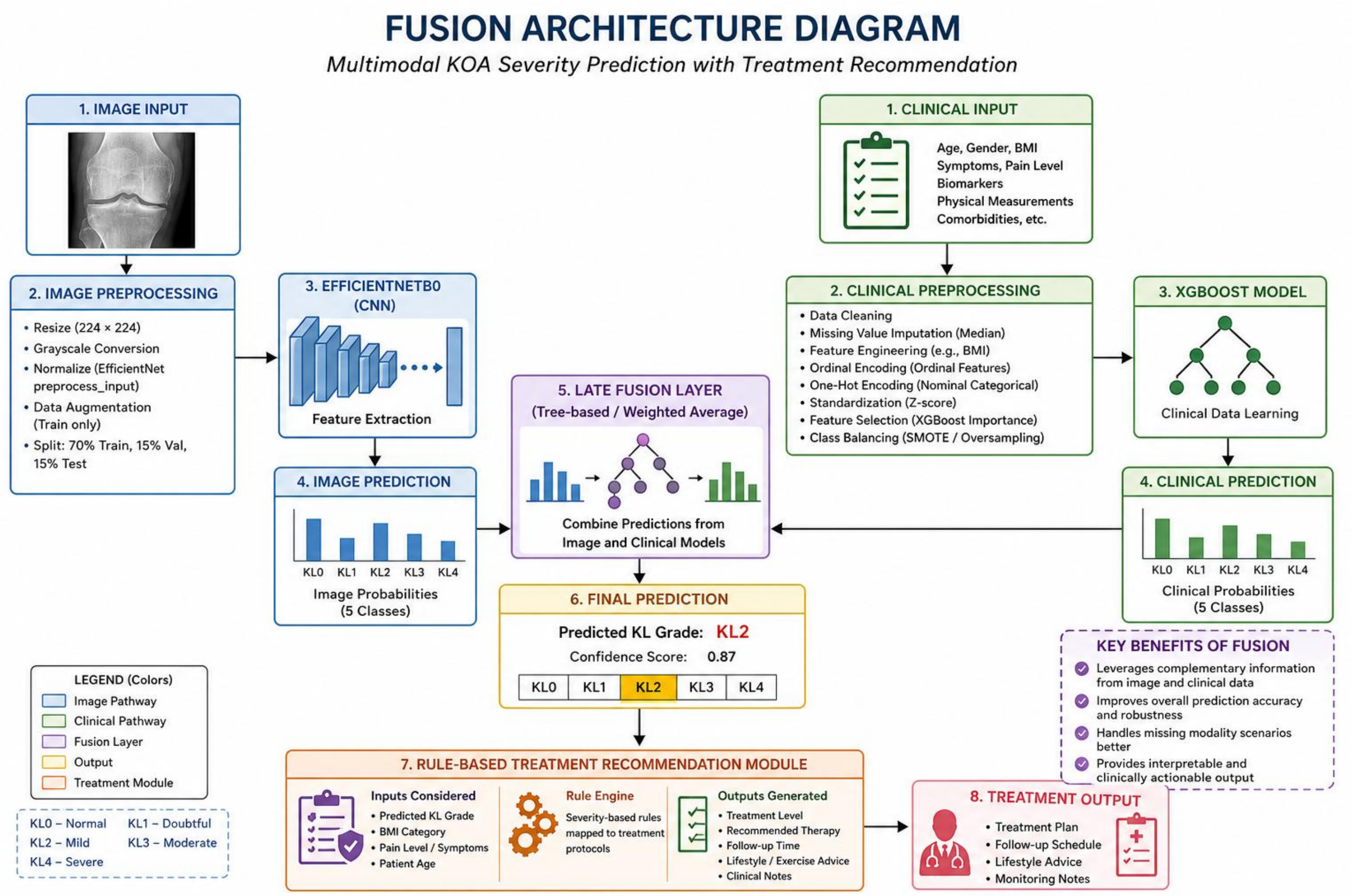

Figure 4 - Component Architecture Diagram

2.1.3. Data Collection

In this module, both imaging and clinical data are used to allow the generation of predictions for the severity of KOA. With these different sources of data being used, there would be a higher chance of having a better understanding of KOA.

For the imaging data, a collection of knee x-ray images were obtained from public sources like the Osteoarthritis Initiative dataset and Kaggle. The images are classified based on the Kellgren–Lawrence grade classification (KL0–KL4), which represent the severity of KOA. Several other datasets have been combined into one, in order to increase generalization and variety of data, while maintaining consistency of class labeling.

Ethical data collection process is employed for collecting clinical data in collaboration with NHSL. Multiple hospital visits were made to collect the data and make sure the dataset is realistic. This dataset contains demographic data (Age, Gender), anthropometric data (Height, Weight, BMI), symptom data (Pain Level, Stiffness, Swelling) and relevant biochemical biomarkers. All patient records were anonymized before any analysis was performed.

2.1.3. Data Preprocessing

Preprocessing was done independently for imaging and structured data to guarantee high-quality, consistent, and compatible data sets that can be used for training our machine learning models. Our data preprocessing process took into consideration the various problems of real data such as missing values, imbalanced data sets, among others.

Preprocessing of Images

For our image data, we had knee X-rays sourced from different publicly available data sets. As all the data sets have varying resolutions, formats, and quality levels, some preprocessing was carried out on the images.

Firstly, all images had been filtered based on file format type (such as JPG, PNG, BMP, WEBP). The corrupted and unreadable images have been detected via img.verify() function and consequently rejected to preserve data integrity.

Afterwards, the each valid image has been loaded and transformed to grayscale) to provide compliance with radiography features. All images had been reshaped to 224 × 224 pixels to fit the EfficientNetB0 model input requirements and been stored in standardized PNG format for more efficient processing and storage.

Class wise data preprocessing procedure has been utilized, where special focus was placed on rare classes (such as KL Grade 4), which had been preprocessed individually. Next, dataset splitting into training (70%), validation (15%), and testing (15%) parts randomly by using random seed for reproducibility purposes.

For the model training, the images were loaded via TensorFlow pipeline with the help of image_dataset_from_directory function and batching/shuffling options applied. Then the EfficientNet-specific preprocessing stage had been performed using preprocess_input.

Preprocessing of Clinical Data

The structured clinical data comprised of demographic, anthropometric attributes, symptoms, and biochemical biomarkers of patients gathered from hospital datasets. Considering the heterogeneous and incomplete character of clinical data, a robust pre-processing mechanism was employed for high-quality data processing purposes.

Firstly, data cleaning involved the removal of redundant attributes like unique IDs, non-significant text data, and inconsistent and/or duplicate entries. Missing values especially in the biomarkers' related features were treated using median imputation to handle outliers and preserve data distribution.

Further, feature engineering techniques were used to extract relevant attributes. For instance, the Body Mass Index (BMI) value was derived from height and weight attributes since BMI is an essential attribute affecting the severity of KOA.

For categorical features, a two-tier approach depending on their characteristics was employed. In the case of ordinal categorical attributes (such as levels of pain and joint stiffness frequencies), they were encoded using the ordinal encoding technique since their order matters. However, in the case of nominal categorical variables (such as gender), they were encoded using one-hot encoding.

In order to maintain numerical stability and equal contribution from all features, continuous features were normalized using z-score scaling technique. It was very crucial in terms of machine learning because it avoids features having larger scale from overwhelming the learning algorithm.

After preprocessing, analysis of feature importance was carried out using XGBoost model which helps to determine which clinical features contribute most towards

KOA severity. Besides improving model interpretation, it was essential to verify the clinical significance of features such as BMI, pain level, and biomarkers among others.

Lastly, since severity levels were not balanced, suitable measures were taken on the training dataset to solve that problem such as creation of artificial data and oversampling. Clinical data was then formatted as feature vectors and fed into the XGBoost model to predict severity level.

2.1.4. Functional Requirements

The software is intended to carry out several fundamental operations, which help achieve KOA severity prediction and treatment recommendations. The following features will enable the system to handle multimodal inputs, make accurate predictions, and present outputs in a clinical manner.

Input Handling and Image Processing

The system will be able to receive knee X-ray images as input data and pre-process them through resizing, normalization, and augmentation to make them ready for the further analysis by the deep learning algorithm.

Processing Structured Clinical Data

The software will receive such input information as the patients' age, gender, height, weight, body mass index (BMI), symptom presence, and biomarker levels and pre-process them through cleaning, encoding, and normalization.

Severity Prediction Based on Imaging Analysis

The software will employ the EfficientNetB0 architecture to learn features from X-ray images and predict the severity grade of the condition based on the KL classification criteria.

Severity Prediction Based on Clinical Parameters

XGBoost algorithm will help analyze the structured clinical data and biomarker levels to make the necessary severity predictions.

Combining Predictions from Models

By using a late fusion technique, the system will be able to combine imaging and clinical predictions and determine the final severity grade of the condition with confidence.

Recommendation of Treatment Plan

The system will recommend the treatment plan in accordance with the predicted severity and individual characteristics like BMI, symptoms, and clinical signs using a rules-based system.

Generation of Output and Data Storage/Retrieval

The system will produce output in the form of predicted KL grade, degree of confidence, treatment level recommendation, period to revisit, lifestyle recommendation, and clinical notes.

Data storage and retrieval will be done safely and securely.

2.1.5. Non-Functional Requirements

Accuracy

Predictive accuracy in the assessment of severity levels among different KOA stages must be ensured. As the decision support system is intended to assist the medical professionals in determining the patient condition, minimizing the error rate in classification and avoiding misclassification of distant severity levels are vital tasks.

Performance and Efficiency

Fast response time should be achieved in terms of prediction generation to make sure that the software could be used in a timely manner. Efficient image processing and data analysis with fast fusion are important to allow users to get the prediction almost instantly. Also, efficient consumption of computing resources in terms of CPU and memory should be assured.

Scalability

It is important to design the software solution that could be easily scaled to work with growing amounts of patients' records and imaging inputs. The scalability of the system is also needed in order to deploy the software in various settings, including the cloud environment.

Reliability

The system must demonstrate reliable performance and be able to generate consistent results upon receiving the same set of input data. The system should not crash or malfunction under usual load, and be robust enough to work with any type of input data sets.

Interpretability

Given that the system is being used in the field of medicine, the need for interpretability is essential. This can be facilitated using approaches like visualization (for instance, Grad-CAM) in the case of images and also analyzing features in clinical data. The inclusion of the rule-based treatment system also improves its interpretability.

Usability

User-friendliness in terms of usability is a quality that the system ought to have. As such, it will have an online interface through which patients' data and pictures can be easily entered and outputs that include severity predictions and treatment suggestions presented effectively to the user.

Security and Privacy

All patients' information is going to be secure in the system since it will be anonymously uploaded. All the protocols used within the web application should meet the standards of protecting patients' information and privacy in healthcare services.

Maintainability

To improve the maintainability of the system, there should be modularity in its design. In this regard, each of its components, including the image model, the clinical model, the fusion process, and the treatment rule, should be independently updatable.

Robustness

The system should be designed to work properly even when the input data is imperfect, faulty, or of poor quality. In other words, even if there are some missing values in clinical data or if there is a slight deterioration in the quality of the images, the system should produce meaningful output or warning messages.

Availability

Because the system operates through a web application, it should always be available. The hosting system should ensure that there is no downtime.

2.2. Commercialization Aspects of the Product

The KOA severity prediction and treatment decision aid system presents high prospects for commercialization due to its applicability, scalability, and feasibility. This is because unlike the case with experimental research systems, this system has been built as a web application that can actually be used. This implies that the system can be accessed from anywhere at any time through the web interface provided.

Through the web application, users, including health care professionals, can upload both knee X-rays and clinical information in order to obtain predictions on the degree of severity and the possible treatments for KOA. Since this is an implemented web application, users can easily access the system from anywhere as long as they have the Internet connection without having to download any special software applications.

Commercially, the product can be provided through a Software as a Service model where hospitals and other medical facilities subscribe to the service provided by the system. Having features like multi modal prediction and treatment decision aid modules makes the system quite valuable from a commercial point of view.

The technology is highly effective in resource-limited conditions where availability of high-end diagnostic equipment such as MRIs might not be present. Through using common X-ray imaging and patient information, the system enables affordable means of screening for and monitoring progression of KOA.

Moreover, thanks to its design as a highly modular framework, the technology can be extended into various commercial possibilities such as:

Integration into EHR systems

Telemedicine solutions

Using data from Internet of Things and wearables

Mobile applications

However, before reaching the commercial market, several issues have to be overcome such as regulatory clearance, compliance regarding data security and privacy, and thorough clinical testing. Furthermore, gaining confidence of healthcare professionals and achieving reliable performance in real-world scenarios are essential steps toward successful commercialization.

In general, development and implementation of the application are key steps to improving commercial success of the system.

2.3. Testing and Implementations

2.3.1. Testing

The development process was done in an iterative approach where tests were performed for purposes of ensuring accuracy, reliability, and feasibility. This was accomplished through the integration of machine learning and deep learning techniques in a pipeline approach, after which extensive tests were conducted on the system.

This was followed by thorough tests that ensured the multimodal system generated accurate KOA severity predictions along with useful treatment recommendations.

Functional Testing

Functional testing was carried out with the aim of determining whether the system was doing what it was supposed to do.

Image Processing Test

Tests performed on the image processing pipeline were aimed at verifying resizing, conversion to grayscale, normalization, as well as corruption detection. The system was successful in ignoring corrupted X-ray images as well as filtering invalid inputs.

Clinical Data Preprocessing Test

The clinical data pre-processing involved performing activities such as missing values removal, encoding (ordinal & one-hot encoding), feature scaling as well as calculating BMI from raw inputs to generate feature vectors.

Severity Prediction Tests

EfficientNetB0 model was tested for accurately classifying KL grades based on X-ray images, and the XGBoost model was validated for clinical data prediction.

Fusion Module Test

The late fusion approach was tested to validate the fusion of predictions from both approaches and produce final outputs for severity with confidence scores.

Recommendation Test

A rule-based system for recommendation was tested on several patients' cases to verify whether there is a correspondence between the recommended treatments and predicted severity levels based on other factors, including patients' BMI and symptoms.

Performance Testing

Performance testing was carried out to determine the efficiency, precision, and scalability of the system.

Model Precision Testing

The model precision testing involved the use of metrics like accuracy, precision, recall, F1-score, and confusion matrix. In addition to the above metrics, for the classification of severity, ordinal mean absolute error (MAE) was used to check the degree of deviation from the true KL grade.

Prediction Execution Time

Prediction execution time per patient was checked for the system to determine whether it can operate effectively in near real-time.

System Resources Usage

Monitoring CPU and memory utilization during prediction was done for effective resource management.

Robustness Testing

The system was subjected to tests involving noisy data, missing data, and low-quality input data.

Integration Testing

Testing was done to ensure that all the components of the system work smoothly with one another.

Data Flow Testing

The data flow of the input data from image data and clinical data, pre-processing, prediction, fusion, and the generation of the output were tested.

Testing of Model Integration

EfficientNetB0 imaging model and XGBoost clinical model were combined and tested to ensure proper integration within the system.

End-to-End Testing

Testing was done to check if the entire system works properly by processing both real and simulated patient data.

System Stability

Testing was done to check the consistency and reliability of the system.

The system was implemented using Python technologies. For building deep learning models, TensorFlow and Keras were used, whereas for clinical prediction models, Scikit-learn and XGBoost were used. Preprocessing of images was done using PIL and TensorFlow pipelines, and for structured data preprocessing, Pandas and NumPy were used.

Due to its modular architecture, separate implementation and testing could be done on all components, which include imaging, clinical prediction, fusion, and treatment recommendation components.

2.3.2. Implementation

The developed KOA severity prediction and treatment support system was built using deep learning and machine learning along with web technologies to provide not only effective analysis but also usability. The main part was developed using the Python language due to a rich set of data preprocessing and modeling tools.

For image processing, TensorFlow and Keras frameworks were used. The EfficientNetB0 network was used with the transfer learning technique. The pre-trained ImageNet weights were used first to train the network on the KOA dataset for further improvement in the domain. Images were processed using PIL and TensorFlow data pipelines with steps like converting them to grayscale, resizing images to 224×224 pixels, normalizing them according to the EfficientNet approach, and loading them in batches using the tf.data API. The training was done in two stages – the initial one was feature extraction, and the second stage was fine-tuning.

Clinical prediction models have been developed using Scikit-learn and XGBoost frameworks, with structured patient data being preprocessed using Pandas and NumPy libraries. Data preprocessing methods include cleaning, dealing with missing values by applying median imputation, creating features such as BMI, encoding categorical variables with ordinal and one-hot methods, and scaling features using standardization. The XGBoost model was used due to its ability to model non-linear associations and better predictive accuracy compared to other models. A late fusion framework was used to combine results obtained from imaging and clinical models. Probabilistic outputs of EfficientNetB0 and XGBoost models were fused using either a weighted average or a rule-based method to generate final KOA severity classification.

Treatment Management Support System is implemented as a rule-based module, in which rules are predefined for each KOA level and can be applied depending on severity level and patient characteristics, such as BMI, pain level, stiffness, and functional limitation. The Treatment Management Support System outputs include recommendation for treatment level, follow-up duration, lifestyle, and clinical notes.

All parts of the system were incorporated into an online application that supported real-time interaction. The front end was designed in such a way that allowed patients to provide their X-ray images and clinical data by filling in forms. The back-end performed the prediction process and other required data manipulations. Application Programming Interfaces (APIs) were used to communicate between the front end and the prediction models.

Modularity was implemented across the whole system, breaking it down into independent blocks: Image Processing Block, Clinical Prediction Block, Fusion Layer Block, and Treatment Recommendation Module. This approach allows for more convenient debugging and testing, as well as possible further modifications in the future, including updating models and improving fusion algorithms.

In general, the implementation provides the necessary compromise between accuracy, efficiency, and scalability.

3. Results & Discussion

3.1. Results

An evaluation of the proposed system for prediction of Knee Osteoarthritis (KOA) severity using image-based and structured clinical data was done.

On the image dataset, EfficientNetB0 performed with a training accuracy of 81.34%, validation accuracy of 78.00% and test accuracy of 80.25%. From the confusion matrix analysis of this model, it performed excellently for high severity cases with KL3 and KL4 with precision at 98.73% and 96.27% recall respectively.

In the clinical case, the proposed XGBoost model showed higher accuracy compared to other machine learning models. It exhibited training accuracy of 88.44%, validation accuracy of 86.88% and test accuracy of 89.00%.

Finally, the system was also able to provide output results based on the severity of KOA and other patient specific information from Treatment Management Support System with rule-based predictions of level of treatment, follow-up period.

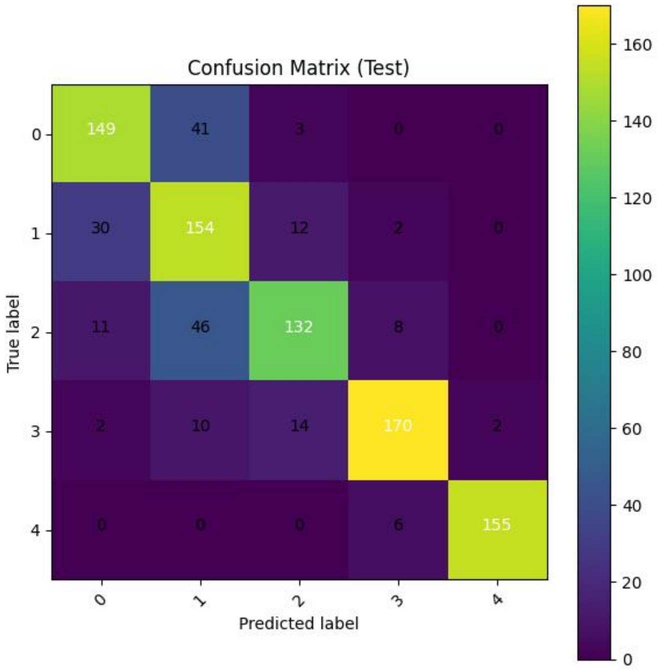

Figure 5 - EfficientNetB0 Model Confussion matrix

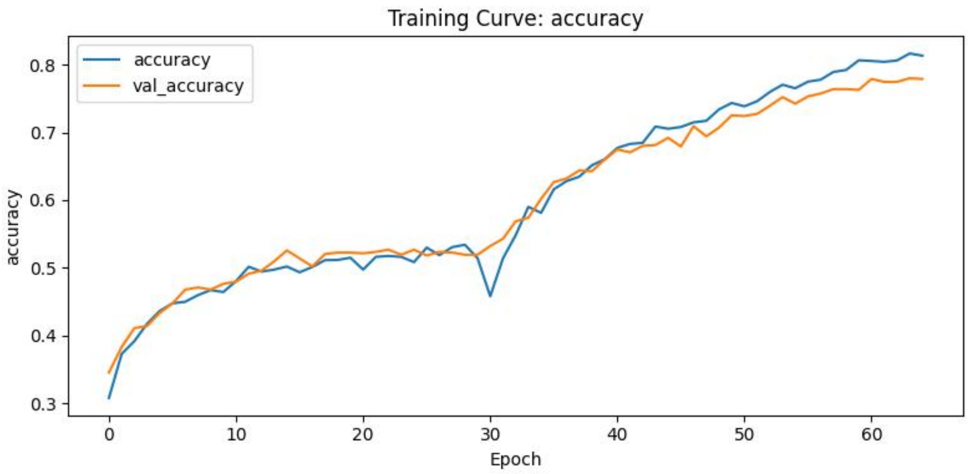

Figure 6 - EfficeintNetB0 Model Accuracy Curve

```
Total features with permutation importance: 104
                             feature  perm_importance  perm_std
3                         pain_score           0.5210  0.034337
7                                 cs           0.0105  0.005679
6                          platelets           0.0090  0.002000
11                                rf           0.0055  0.004153
10                               esr           0.0050  0.003162
9                                crp           0.0040  0.003000
8                        cholesterol           0.0025  0.005123
17   physical_activity_level_Moderate           0.0025  0.002500
2                             weight           0.0025  0.003354
4                                fbs           0.0020  0.005099
```

Figure 7 - Feature importance analysis of the clinical model (XGBoost)

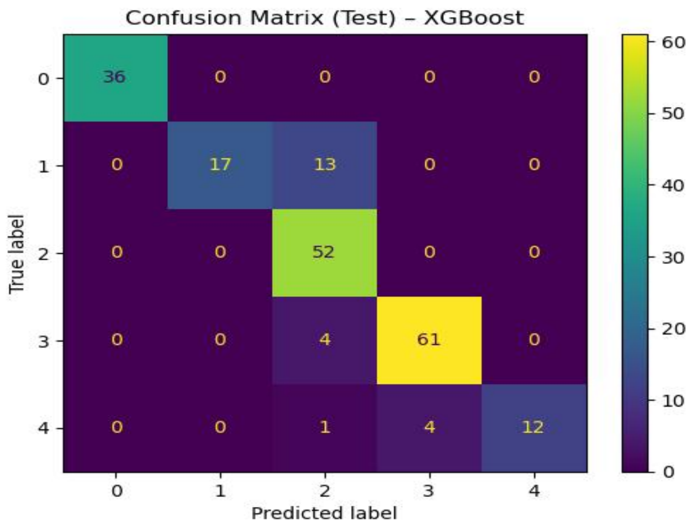

Figure 8 - Confusion matrix of the XGBoost clinical model

3.2. Research Findings

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As observed from the experimental findings, there are several significant results. First, the clinical model was more effective than the imaging model in terms of overall accuracy, implying the presence of robust predictive features in the clinical and biomarker data for KOA severity. Parameters such as BMI, pain, and biomarkers were highly significant for predicting KOA.

Secondly, the imaging model performed excellently in classifying the high severity classes such as KL3 and KL4 owing to the presence of distinctive features of such classes compared to early stages.

In addition, the misclassification error occurred between the neighboring severity classes (e.g., KL1 and KL2). Consequently, the predictions by the models did not differ greatly from the actual severity class.

Finally, it was possible to translate the predictions into meaningful suggestions through the use of a rule-based system.

3.3. Discussion

From the results, we can conclude about the advantages and disadvantages of using imaging and clinical methods for predicting the severity of KOA. The higher quality of the clinical model may be explained by the fact that in this case, there is the possibility of applying the model to a structured dataset and working with clear markers of the development of the disease. On the other hand, it is harder for the model to work with imaging data since it should understand the visual patterns. Moreover, in case of early-stage KOA, there may not be enough clear signs for that.

High efficiency of classifying KOA at the stages with high severity indicates that deep neural networks, such as EfficientNetB0, can easily identify radiographic anomalies. Nevertheless, low results in distinguishing adjacent classes show that multi-class classification is indeed difficult for deep neural networks due to the similarity of certain stages.

Thus, we can see that the multimodal fusion method should be applied to increase the precision of classification

4. 4.Summary of Individual Contribution –Perera B.B.A.R.(IT22606792)

For this module, the project focused on developing a predictor of KOA severity based on imaging and clinical data, together with treatment recommendations. I contributed to both the creation of the deep learning/ML algorithms and the rules, presenting a comprehensive package from a front-end web application to its back end.

One major highlight was developing the predictor based on imaging. Preprocessing included validating the data, converting them to gray scale, resizing, normalization, and then using them to train the network. Transfer learning and fine tuning were applied to make use of EfficientNetB0 to predict the severity of KOA based on the Kellgren-Lawrence scale.

Another predictor was created using clinical data. Data preprocessing included dealing with missing values, one-hot encoding, feature scaling, BMI calculation, and calculating feature importance.
Also noteworthy is the incorporation of a multimodal fusion approach by merging the results from the two models to create better predictions and robustness using a late fusion method.

Rule-based Treatment Management Support System refers to the creation of a system to derive treatment decisions from predicted values. This part of the assignment includes generating output results such as the treatment level, follow-up time period, lifestyle, and clinical notes as guided by specific patients' information.

This contribution involves the development of the web application, which entails the development of the frontend and backend of the software. For the frontend development, a web-based form that will take X-rays as well as other forms of clinical data as input data was created. For the backend, there was a need to develop software for processing the input data, inference of the models, and integration of the imaging, clinical, fusion, and treatment management systems. API services were developed to integrate both the frontend and backend systems seamlessly.

Overall, this part involved the development of an end-to-end process starting from data pre-processing up to web applications and full-scale modeling.

5. Conclusion

In particular, the developed system succeeded in creating and evaluating a multimodal model for automated prediction of Knee Osteoarthritis (KOA) severity and assistance in treatment decisions using both radiographic imaging and structured data from patients' medical history. The multimodal model included two submodels based on a convolutional neural network with EfficientNetB0 architecture for analyzing imaging data and XGBoost model for analyzing clinical data, after which late fusion was implemented.

Based on the obtained experimental results, one can draw a conclusion about the advantage of using the clinical model because of its greater predictive power owing to a high level of structure and informativeness of used features. On the other hand, the importance of using the imaging model has been highlighted, since it provided new insights into the structure changes occurring in the knee joint. In addition, the obtained results showed that the misclassification between neighboring Kl grades occurs due to difficulties with the classification of such cases.

Another significant achievement of this work lies in the creation of the rule-based Treatment Management Support System, which translates the predicted severity into the treatment level recommendation, treatment period, and other severity-based recommendations. In contrast to the black box approach used in the prediction phase, the output provided by the Treatment Management module is clear and interpretable, which includes treatment level, time needed for follow-up, recommended lifestyle changes, and clinical notes. As a result, such output adds value by connecting the prediction step and the actionable clinical decision.

Moreover, the creation of the application and its availability as a web-based platform indicate its applicability. The capability of processing imaging and clinical data and delivering results in real time allows making the system scalable and accessible for any environment.

To conclude, this study contributes to the field of intelligent healthcare systems by developing the decision support system for assessing the severity of KOA and recommending a treatment course that can be easily integrated into the clinical routine. Not only did this system achieve high accuracy in prediction by using multimodal data, but it also helped solve the issue of clinical interpretability by providing interpretable recommendations.

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